

Sparse Feature Selection from Low-Montage Resting-State EEG for tDCS Treatment Remission Prediction in Adults with Major Depressive Disorder

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ABSTRACT

We evaluated whether sparse feature selection from 4-channel resting-state EEG could identify remission predictors in a home-based tDCS trial for major depression. Across 150 configurations of a hybrid CNN-LSTM and linear SVM, we varied window length (2–10 s) and feature budget (1–70) under leave-one-participant-out cross-validation with SMOTE.

The strongest hybrid configuration (6 s windows, one feature per fold) achieved patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, and MCC 0.571 over 21 held-out participants. A permutation test confirmed above-chance discrimination (raw $p = 0.008$; 2-family Bonferroni-corrected $p_{\text{corr}} = 0.016$), although significance does not survive correction for the full 150-configuration sweep. The best SVM (8 s, 30 features) reached ROC-AUC 0.510 ($p = 0.469$, uncorrected). The hybrid model exhibited extreme sparsity: only 7 unique features appeared across folds, with modest overlap (mean Jaccard 0.190) but perfect effect-direction consistency among recurrent features.

These findings suggest that patient-level discrimination signals are recoverable from highly sparse representations of low-montage EEG. The recurrent features centred on temporal spectral entropy and frontal-temporal difference measures, converging with prior analyses in the same cohort. The results support low-montage resting-state EEG as a basis for pretreatment stratification research, pending independent validation.

1. Introduction

Major depressive disorder (MDD) remains a leading cause of global disability and is characterized by marked heterogeneity in clinical presentation, neurobiology, and treatment response. Despite the availability of pharmacological and psychotherapeutic interventions, a substantial proportion of patients fail to achieve remission, motivating increasing interest in neuromodulation approaches such as transcranial direct current stimulation (tDCS) (Woodham et al., 2021). While randomized controlled trials and recent large-scale studies support the feasibility and clinical potential of tDCS, including fully remote, home-based delivery, treatment outcomes remain variable at the individual level (Hampsey et al., 2026; Woodham et al., 2025).

1.1. Clinical significance of low-density EEG

Home-based delivery has moved from concept to clinical trial: a fully remote phase 2 randomized sham-controlled study demonstrated home-based tDCS in major depressive disorder (Woodham et al., 2025), and qualitative follow-up work found the approach acceptable to participants (Rimmer et al., 2024). These advances make pretreatment stratification a practical clinical problem. A

central challenge is pretreatment stratification: identifying which patients are more likely to achieve remission before treatment begins. Electroencephalography (EEG) is a potential modality for this purpose due to its portability, low cost, and high temporal resolution. A 4-channel low-montage configuration (AF7, AF8, TP9, TP10) is particularly attractive for home-based settings: the headset is lightweight, self-applicable by patients with minimal training, and substantially less expensive than high-density laboratory systems. However, this reduced channel count necessarily trades off spatial coverage (capturing only bilateral frontal and temporal regions) and signal richness against practical deployability. A key open question is whether the limited spatial sampling of a 4-channel montage retains sufficient information for clinically useful treatment-response prediction, or whether the signal loss relative to 32- or 64-channel systems is prohibitive. The present study directly addresses this tension by testing whether rigorous feature engineering and sparse selection can recover predictive signal from the same low-density setup used in the parent home-based trial.

The literature on EEG biomarkers in major depressive disorder is encouraging though methodologically uneven. Reviews have highlighted spectral, asymmetry, and entropy signatures as potentially informative marker families (Leuchter et al., 2010; Olbrich and Arns, 2013; Olbrich et al., 2015), but also emphasize the need for more robust treatment-response models and better translational discipline (Fu et al., 2019; Widge et al., 2019).

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Meta-analytic work on EEG-based treatment prediction reaches a similar conclusion: useful signal is present, but translation depends on stronger methodology and standardization (Cohen et al., 2023; Simmatis et al., 2023; Watts et al., 2022). Machine learning studies have extended this into antidepressant and neuromodulation response prediction while exposing recurrent concerns about external generalization, feature stability, and leakage-safe selection (Al-Kaysi et al., 2017; Jaworska et al., 2019; Oakley et al., 2022). Leakage-specific critiques, in particular, make the underlying design problem explicit (Shen et al., 2025; Shim et al., 2021). In relatively small neuromodulation studies, rich models risk unstable feature selection and inflated performance estimates, making nested evaluation and within-fold processing central design issues rather than implementation details (Shim et al., 2021; Widge et al., 2019). Emerging response-prediction work continues to show promise while exposing the same methodological fragility (Dagnino et al., 2023; Guo et al., 2024; Ulrich et al., 2025).

Within the same home-based 4-channel MDD program analysed here, power spectral density (PSD)-based and PLV-based analyses have observed that the baseline EEG carries treatment-response information across multiple signal representations (Moncy et al., 2024; Xiao et al., 2025b). PSD-based deep learning approaches identified temporoparietal spectral features, with TP10 theta-alpha power as informative for treatment outcome prediction (Moncy et al., 2024), while connectivity-based analyses highlighted frontal-temporal and temporoparietal network patterns (Xiao et al., 2025b). Together, these findings suggest that response-related signal is present and reproducible across feature families, but remains poorly characterized in terms of interpretability, stability, and minimal feature representation.

The present study builds directly on this context and uses the same baseline EEG cohort as our prior PSD- and connectivity-based analyses (Moncy et al., 2024; Xiao et al., 2025b), but addresses a distinct question. Rather than optimising predictive performance within a predefined feature family, we test whether a strictly leakage-controlled, foldwise sparse feature-selection pipeline can identify a small, recurrent, and interpretable feature set under participant-level validation. We compared two classifiers over the same feature-selection sweep to test whether low-montage resting-state EEG could recover a small recurring set of features associated with tDCS remission in adults with MDD. The analysis prioritized parsimony, interpretability, and leakage-resistant participant-level validation. Our working hypothesis was that useful patient-level discrimination could be achieved with a small per-fold feature budget and that the most recurrent features would show stronger effect-direction consistency than hard-set overlap. This approach reframes the problem from high-dimensional modelling toward a minimal, interpretable signal identification task.

2. Methods

2.1. Dataset and preprocessing

The analysis used pretreatment EEG from 21 adults with major depressive disorder enrolled in a fully remote, multisite, double-blind, randomized sham-controlled home-based tDCS trial (Woodham et al., 2025). The analyzed cohort contained 7 remission cases and 14 non-remission cases in the patient-level outcome labels, with 18 female participants and a mean age of 37.1 ± 9.7 years. Table 1 summarizes the cohort, acquisition, and sweep configuration.

Baseline recordings were collected during eyes-closed rest with a 4-channel montage (AF7, AF8, TP9, TP10) sampled at 256 Hz. Figure 1 shows the recording sites together with the corresponding headset placement. The available study exports comprised 10 s segments with 2,560 samples per channel, and the underlying recording sessions were stored either as a single 10 min file or as two 5 min files depending on participant. Across the cohort, the export set contained 1,203 windows, including 393 remission-labeled and 810 non-remission-labeled windows. This acquisition setup uses a limited frontal-temporal montage with clear portable-deployment relevance (Miranda et al., 2019). Four channels were chosen as a deliberate practical compromise: enough to cover bilateral frontal (AF7/AF8) and temporal (TP9/TP10) regions while remaining simple enough for unsupervised home self-application without technician assistance. This configuration omits central, parietal, and occipital sites, sacrificing spatial resolution and precluding the analysis of well-studied posterior markers (e.g., parietal alpha asymmetry). The trade-off is explicitly motivated by the target use case: a fully remote, home-based tDCS-monitoring setting in which a 19- or 32-electrode research cap is infeasible. The montage also constrained the downstream feature space toward frontal, temporal, cross-hemispheric, and frontal-temporal interaction measures.

For the present sweep, the processing configuration applied participant-level per-channel demeaning before upsampling, followed by a fourth-order Butterworth low-pass filter at 60 Hz, a downsampling step back to 256 Hz, and zero-overlap window slicing. The 60 Hz cutoff was chosen to retain the full gamma band while providing attenuation above the Nyquist frequency of the original 256 Hz recording; a notch filter at 50 Hz (UK mains frequency) was not applied because the low-pass rolloff already provides substantial attenuation. The upsample/downsample pipeline preserved signal fidelity while allowing consistent filter implementation across all window sizes. The sweep then compared non-overlapping analysis windows of 2, 4, 6, 8, and 10 s, so the model inputs included both the native 10 s exports and shorter subdivisions derived from the same recordings. These choices materially affect downstream EEG feature distributions, especially in low-density or portable pipelines (Mahato et al., 2020; Miranda et al., 2019; Siuly

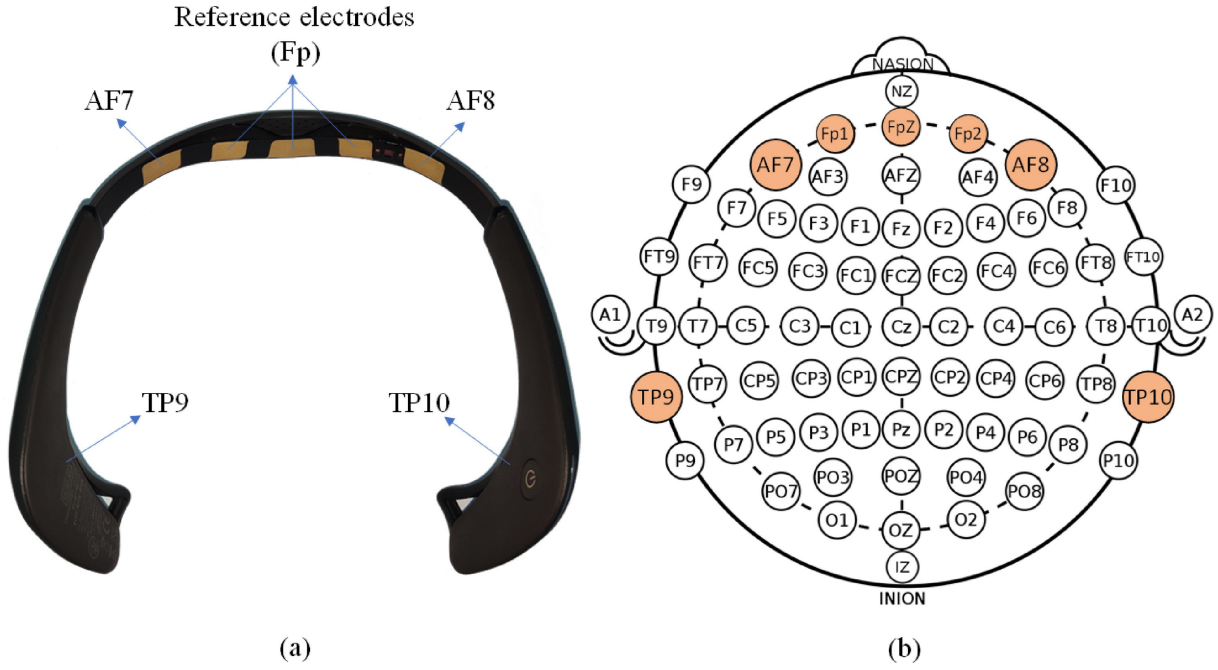


Figure 1: Four-channel low-montage EEG setup corresponding to the recording configuration used in the study, showing the AF7, AF8, TP9, and TP10 recording sites together with the corresponding headset placement.

et al., 2017). They also intersect with standardization and signal-quality concerns in wearable EEG settings (BABILONI et al., 2012; Simmatis et al., 2023).

2.2. Feature extraction

Feature extraction was performed on each preprocessed window. With all four channels available, the extractor generated 247 EEG-derived measures per window spanning channel-local descriptors, regional asymmetry terms, frontal-temporal interaction terms, synchrony summaries, and a small set of TP10-specific derived measures. Each channel contributed 39 local measures, comprising 21 spectral descriptors, 12 time-domain and Hjorth descriptors, and 6 entropy or nonlinear complexity descriptors, for 156 channel-local measures overall. The remaining 91 measures summarized asymmetry, frontal-temporal balance, cross-channel synchrony, and TP10-focused derived descriptors. This broad feature basis allowed multiple interpretable feature families to be tested within a leakage-safe sparse-selection framework rather than pre-committing the analysis to a single biomarker hypothesis (Shen et al., 2025; Zhang et al., 2023). Table 2 summarizes the extracted feature families and their computational rationale.

For each of AF7, AF8, TP9, and TP10, we computed 39 channel-local features. Spectral features were estimated from Welch power spectral densities (Welch, 1967) and included absolute band power in the delta, theta, alpha, beta, and gamma ranges, total power, relative band power normalized by total power, frequency-weighted power, spectral edge frequencies at 90% and 95%, and PSD summary statistics (mean, standard deviation, maximum,

minimum, median, skewness, kurtosis). These measures were included because spectral content remains one of the most widely used and interpretable feature families in depression EEG work (Mahato et al., 2020; Mumtaz et al., 2017; Olbrich and Arns, 2013). Temporal-channel spectral content has also appeared repeatedly in pretreatment response studies (Jaworska et al., 2018).

Time-domain descriptors were also computed for each channel. These included mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, and mean absolute deviation, together with Hjorth activity, mobility, and complexity (Hjorth, 1970; Zhang et al., 2023). These features summarize amplitude dispersion, local oscillatory rate, signal roughness, and short-window morphology. They are computationally light and well suited to low-montage resting-state EEG, where clinically useful features must remain interpretable and robust under short windows.

Entropy and nonlinear complexity measures were computed per channel to capture irregularity beyond simple power summaries. The extractor included sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis. Complexity measures were calculated on a demeaned, standardized version of each signal, with long windows subsampled for numerical stability. These descriptors were retained because depression EEG studies have repeatedly implicated entropy, variability, and broader complexity structure (Acharya et al., 2015; Faust et al., 2014; Puthankattil and Joseph, 2012). That pattern also appears in pretreatment response and large-scale resting-state evaluations, even when the exact direction of effect depends

Table 1

Cohort, acquisition, and sweep configuration summary. Trial descriptors derive from the parent home-based tDCS study (Woodham et al., 2025); acquisition totals and sweep settings derive from the analyzed EEG exports and current experiment configuration.

Characteristic	Value
Parent trial and cohort descriptors	
Parent trial	Fully remote, multisite, sham-controlled home-based tDCS trial for MDD (Woodham et al., 2025)
Participants	21
Outcome groups	7 remission, 14 non-remission
Sex (female/male)	18/3
Age (years, mean $\pm$ SD)	37.1 $\pm$ 9.7
EEG acquisition and export descriptors	
Rest condition	Eyes-closed resting state
EEG montage	4 channels (AF7, AF8, TP9, TP10)
Channel coverage	Frontal (AF7, AF8), temporal (TP9, TP10)
Sampling rate	256 Hz
Recording structure	10 min total; single file or two 5 min files
Available export segment length	10 s (2,560 samples/channel)
Total exported windows	1,203
Remission windows	393 (32.7%)
Non-remission windows	810 (67.3%)
Class ratio	2.06:1 (non-remission:remission)
Current sweep methodology settings	
Sweep window sizes	2, 4, 6, 8, 10 s
Inner-k sweep	1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70
Feature selector	SelectKBest (f -classif)
Training adjustments	LOPO-group equalization + SMOTE
Validation	Leave-one-participant-out
Evaluation level	Patient-level aggregation

Table 2

Summary of extracted EEG feature families per analysis window.

Feature family	Measures	Count	Why included
Channel-local spectral	Absolute and relative delta, theta, alpha, beta, and gamma power; total power; frequency-weighted power; SEF90 and SEF95; PSD mean, standard deviation, maximum, minimum, median, skewness, and kurtosis	21/channel (84 total)	Canonical resting-state EEG descriptors with clear band-level interpretation and strong precedent in depression EEG
Channel-local time domain and Hjorth	Mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, mean absolute deviation, Hjorth activity, mobility, and complexity	12/channel (48 total)	Captures amplitude dispersion, oscillatory rate, roughness, and short-window morphology with low computational cost
Channel-local entropy and nonlinear complexity	Sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis	6/channel (24 total)	Captures irregularity and complexity beyond band-power summaries
Band-power asymmetry and frontal-temporal interactions	Frontal and temporal asymmetry, hemispheric asymmetry, ipsilateral frontal-temporal ratios and normalized differences, and diagonal AF7-TP10 and AF8-TP9 interactions across frequency bands	70 total	Tests laterality and frontal-temporal balance within the four-channel montage
Synchrony and coupling	Pearson correlation and maximum normalized cross-correlation over six anatomical channel pairs, plus legacy cross-correlation summaries	18 total	Summarizes short-window coupling and cross-channel coordination
TP10-specific derived descriptors	Peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term	3 total	Targets concentrated temporal spectral structure with straightforward interpretation
Total EEG-derived features	Combined feature space per window	247 total	Broad hypothesis space for leakage-safe sparse selection

on cohort, montage, and temporal scale (Jaworska et al., 2018; Lord and Allen, 2023; Zhao et al., 2022).

The four-channel montage also supports a range of regional interaction features derived from cached band-power estimates. Traditional frontal asymmetry (AF7/AF8) and temporal asymmetry (TP9/TP10) were computed for each frequency band in two forms: a log-ratio index,

$$AI = \log\left(\frac{P_R}{P_L}\right),$$

and a normalized difference,

$$ND = \frac{P_R - P_L}{P_R + P_L}.$$

Using the same band-power cache, we also computed hemispheric asymmetry by averaging AF7+TP9 and AF8+TP10 within each hemisphere, ipsilateral frontal-temporal ratios and normalized differences on the left and right hemispheres, and diagonal frontal-temporal interactions for AF7-TP10 and AF8-TP9. These measures were included because reduced-channel depression EEG studies repeatedly point to laterality, temporal involvement, and frontal-temporal balance rather than to a single isolated electrode (Ghiasi et al., 2021; Mahato et al., 2020; Monni et al., 2022).

Pairwise synchrony features were added for six anatomically meaningful channel pairs: AF7-AF8, TP9-TP10, AF7-TP9, AF8-TP10, AF7-TP10, and AF8-TP9. For each pair we computed maximum normalized cross-correlation and Pearson correlation, and the extractor also retained a legacy set of raw cross-correlation summaries across all channel pairs. These features were intended to capture short-window synchrony and coupling patterns that have been emphasized in resting-state depression EEG and neuromodulation response work (Ghiasi et al., 2021; Ulrich et al., 2025; Zhao et al., 2022).

Finally, the extractor included three exploratory TP10-specific derived descriptors: peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term. These summarize whether the TP10 spectrum contains a sharp dominant peak, how concentrated that peak is relative to total power, and whether that concentration covaries with ipsilateral frontal-temporal beta balance. They were included because temporal-channel spectral content has recurrently appeared in low-montage depression EEG analyses, while still remaining interpretable enough to survive foldwise sparse selection if informative (Mahato et al., 2020; Mumtaz et al., 2017).

2.3. Sweep design and validation

The experiment sweep compared two model families over the same feature-selection protocol: an advanced hybrid 1D CNN-LSTM and a linear SVM. All runs used sequential window ordering, SelectKBest with ANOVA F statistics, leave-one-participant-out evaluation,

leave-one-participant-out training-group equalization, and SMOTE within training folds (Chawla et al., 2002). The sweep varied the per-fold feature-selection budget (inner- k) across 15 values from 1 to 70 and repeated the search across 5 window sizes, yielding 150 successful runs in total (75 per model).

This design matters for interpretation. Leave-one-participant-out evaluation remained subject independent throughout, so no participant contributed both training and test windows to the same fold. Feature selection, participant-group equalization, and synthetic oversampling were all restricted to the corresponding training participants, which is important because leakage at these steps can materially inflate reported biomarker performance (Shim et al., 2021). The inner- k parameter controlled how many features were selected inside each training fold. The feature-selection stability panels in the study describe shared pipeline behavior because both classifiers consumed the same selected features for matched sweep settings.

2.4. Models

Both classifiers operated on the fold-selected handcrafted feature vectors produced by the current inner- k setting. Patient-level outputs were derived by averaging held-out window probabilities within each participant before computing ROC-AUC and the other clinical metrics. Table 3 summarizes the model architecture and training setup used in the sweep.

The SVM baseline used a linear kernel with $C=1.0$ and probability output enabled. When SMOTE was active inside the training folds, class weighting was not applied in parallel, so synthetic oversampling and explicit class weighting were not used simultaneously.

The hybrid model treated the selected feature vector as a learned one-dimensional representation before recurrent modeling. In the best-performing 6 s, inner- $k=1$ run, the input features were first projected to a 128-dimensional embedding with normalization and dropout, then mapped into a 64-position latent representation for three convolutional blocks. These blocks used channel progressions of [64,64], [128,128], and [256,128] with kernel sizes [5,3], [3,3], and [3,1], together with batch normalization, max pooling, and dropout after each stage.

Sequence modeling then used two bidirectional LSTM layers with 128 and 64 hidden units, followed by eight-head self-attention with residual layer normalization and dropout. The recurrent summary was globally pooled and concatenated with pooled multi-scale CNN outputs through feature-pyramid fusion before two dense blocks of 256 and 128 units and a final two-logit output layer. This architecture distributes representational capacity across complementary stages: the projection and convolutional front end compresses and summarizes the selected features, bidirectional LSTMs provide sequential modeling, self-attention reweights informative positions, and feature-pyramid fusion retains multi-scale representations.

Resting-State EEG Processing and Evaluation Pipeline

In each fold, one participant is held out once; rebalancing and feature selection are performed only on the remaining training participants.

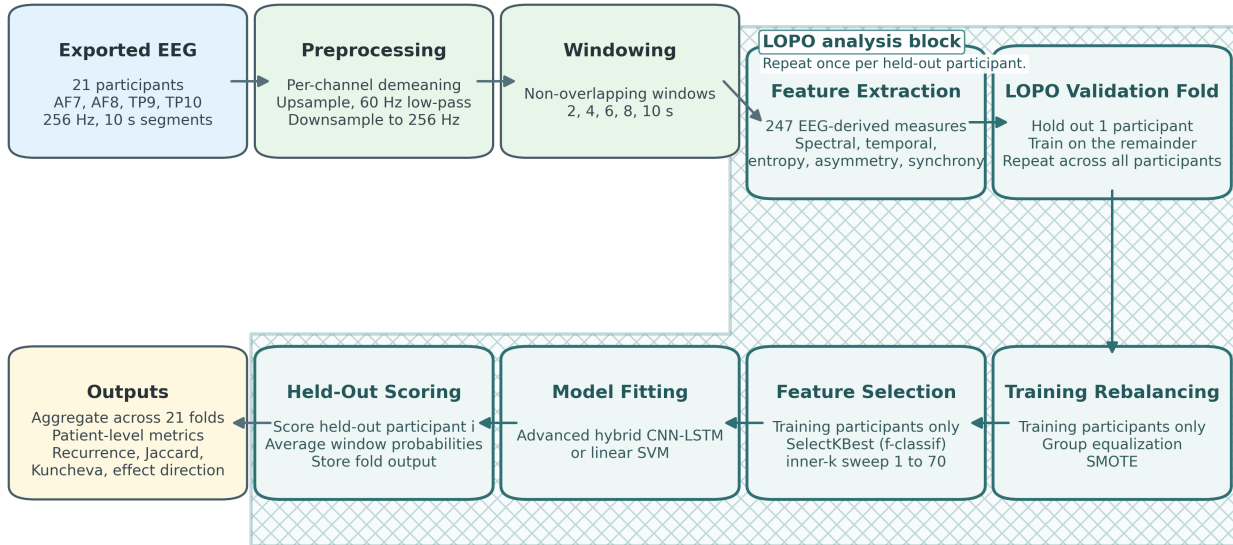


Figure 2: End-to-end analysis flow used in the study. In each leave-one-participant-out fold, one participant was held out once, and participant-group equalization, SMOTE, and feature selection were all applied only within the remaining training participants before held-out scoring and biomarker summarization.

Training used AdamW, cosine warm restarts, label smoothing, gradient clipping, mixed precision, and early stopping with best state restore.

The hybrid model was trained with full-batch gradient descent per LOPO fold. The configured batch-size ceiling of 8192 exceeds the largest per-fold dataset (approximately 5,000 windows at 2 s) and is automatically reduced to the training-set size by the batch-resolution logic (effective batch approximately 2,300 windows at 6 s). Full-batch was chosen because the architecture contains BatchNorm1d after each convolutional block; with per-fold datasets of only 2,000–5,000 windows, mini-batches would produce high-variance normalization statistics. Full-batch accumulates stable, population-level statistics from the entire training fold. The loss of stochastic gradient noise is mitigated by the 21 independent LOPO splits, which assess generalization through cross-validation structure rather than through a single optimization trajectory.

The SVM was selected as a well-characterized linear baseline. The hybrid CNN-LSTM was chosen because its convolutional, recurrent, and attention-based stages can learn interactions among a small set of selected features. All architectural hyperparameters (layer sizes, kernel sizes, attention heads, learning rate, dropout schedule, optimizer settings) were fixed a priori from a standard configuration used across multiple model families in this pipeline; none were tuned on the current cohort. The inner-k and window-size sweep represents the only structured hyperparameter exploration in this study. At inner-k=1, SMOTE operates in a 1-dimensional feature space where

the standard rationale for synthetic oversampling does not fully apply; the no-SMOTE ablation in Section 3.2a quantifies this sensitivity.

2.5. Outcome metrics and feature summaries

The primary endpoint was patient-level ROC-AUC, computed from participant probabilities aggregated over held-out windows. Secondary patient-level metrics included PR-AUC, balanced accuracy, accuracy, F1, MCC, precision, recall, and specificity. The pipeline also reported bootstrap confidence intervals and a permutation test for the patient-level ROC-AUC summary. All main study claims use patient-level metrics. Window-level scores appear only as intermediate outputs.

To characterize biomarker behavior across leave-one-participant-out folds, we used several post hoc summaries: mean pairwise Jaccard overlap, Kuncheva index, unique feature count, top-feature share, class-conditional selection deltas, and effect-direction consistency. These summaries are descriptive. They characterize recurrence and directionality across folds and do not support causal interpretation. Low overlap can coexist with recurring features that retain consistent class-separation direction when selected (Olbrich and Arns, 2013; Olbrich et al., 2015; Simmatis et al., 2023).

Table 3

Model architecture and training summary for the two evaluated classifier families.

Component	Configuration
Input to both models	Fold-selected handcrafted feature vector from the current <i>inner-k</i> setting. Patient-level predictions were computed by averaging held-out window probabilities within each participant.
SVM baseline	Linear-kernel SVM with $C=1.0$ and probability estimates enabled. When SMOTE was active in the training folds, class weighting was disabled to avoid double balancing.
Hybrid input embedding	Linear projection from the selected feature vector to 128 latent units, followed by normalization, a nonlinear activation, dropout 0.1, and remapping to a learned one-dimensional sequence of length 64.
CNN trunk	Three 1D convolutional blocks with filter progressions [64,64], [128,128], and [256,128]; kernel sizes [5,3], [3,3], and [3,1]; max-pooling factors 2, 2, and 1; batch normalization; and dropout [0.1, 0.2, 0.3].
Recurrent stage	Two bidirectional LSTM layers with 128 and 64 hidden units followed the CNN trunk.
Attention and fusion	Eight-head self-attention with dropout 0.1 and residual layer normalization, followed by global mean pooling. Multi-scale pooled CNN features were concatenated through feature-pyramid fusion.
Dense head	Two dense blocks of 256 and 128 units with batch normalization, nonlinear activations, and dropout 0.3 and 0.4, followed by a two-logit output layer.
Optimization	AdamW with learning rate 0.001 and weight decay 0.01; cosine warm restarts to a minimum learning rate of 10^{-6} ; label smoothing 0.05; gradient clipping at 1.0; mixed precision; maximum 100 epochs; early stopping patience 5 with best-state restore.
Training strategy and batch sizing	
Training regime	Full-batch gradient descent per LOPO fold. The batch-size ceiling of 8192 is a guard value, not a tuned hyperparameter: it was set to exceed the largest per-fold dataset in the sweep ($\approx 5,000$ windows at 2 s) and is automatically reduced to the training-set size by the batch-resolution logic (effective batch $\approx 2,300$ windows at 6 s). Full-batch was chosen deliberately because the architecture contains BatchNorm1d layers after every convolutional block; with per-fold datasets of only 2,000–5,000 windows, conventional mini-batch sizes (32–128) would produce high-variance normalization statistics that could encode participant identity rather than population-level features. Full-batch accumulation yields stable, reproducible statistics from the entire training fold and is tractable because each LOPO fold is trained independently on a single GPU.

3. Results

3.1. Sweep-level comparison

Across the 150 successful runs (75 hybrid, 75 SVM), the hybrid model outperformed the SVM baseline on patient-level discrimination metrics over most of the sweep. The performance advantage was visible when averaging over window sizes (Figure 3) and when averaging over *inner-k* settings (Figure 4). The best individual hybrid run occurred at 6 s windows with *inner-k*=1, whereas the best SVM run occurred at 8 s windows with *inner-k*=30.

To assess statistical robustness across the structured sweep, we applied three correction tiers to the permutation p -values. Treating the sweep as two model-family comparisons (the primary independent factor) yields Bonferroni-corrected $p_{\text{corr}} = 0.016$, confirming that the result survives at $\alpha = 0.05$. A max-T permutation correction (10,000 iterations with shared label permutations across all 150 configurations, accounting for inter-configuration correlations) yields $p_{\text{corr}} = 0.302$; the null maximum distribution has mean 0.768 and 95th percentile 0.908, placing the observed maximum of 0.816 within the range expected under random label permutations. The full-sweep Bonferroni ($p_{\text{corr}} = 1.00$) is mechanically uninformative given the 150-test multiplier and is reported for completeness. We adopt the structured 2-family correction as the primary corrected p -value because the window-size and *inner-k* factors represent exploratory

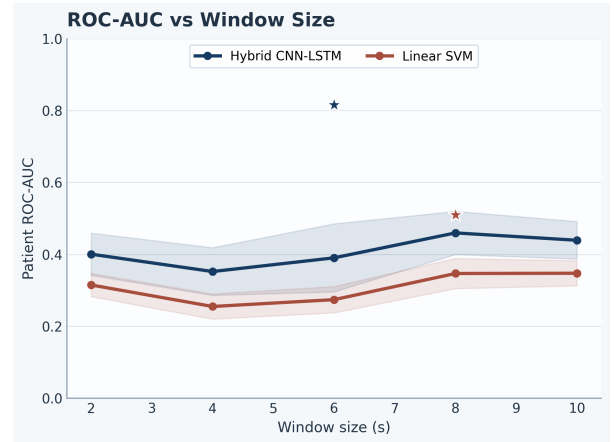


Figure 3: Patient-level ROC-AUC by window size across the sweep. Stars mark the best individual hybrid and SVM runs.

levels of a single hypothesis rather than independent comparisons. Across the sweep, 2 of 150 runs (1.3%) yielded raw $p < 0.05$, and 88% of runs had $p \geq 0.50$, consistent with a null-dominated sweep from which the best hybrid configuration meaningfully separates.

Figure 4 shows that the relationship between ROC-AUC and *inner-k* differed across the two model families in the low-budget region. The hybrid model retained clearly stronger ROC-AUC across the sparsest budgets and achieved the highest individual ROC-AUC in the entire

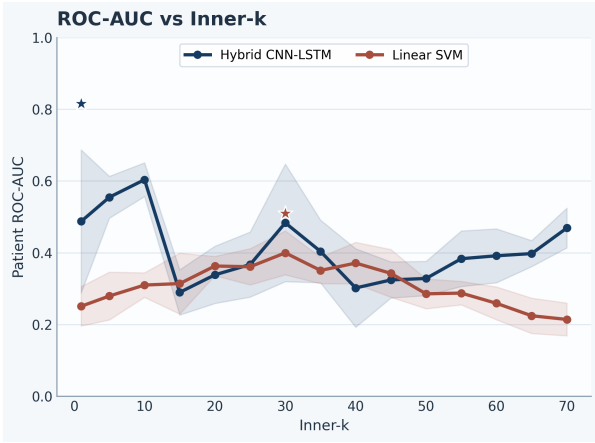


Figure 4: Patient-level ROC-AUC by inner-k across the sweep. The hybrid model retained the strongest discrimination in the sparsest budget region.

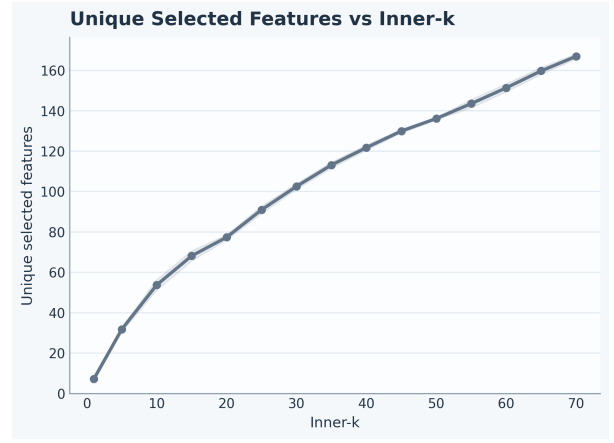


Figure 6: Unique selected features by inner-k. Larger per-fold budgets produced markedly larger discovered feature pools.

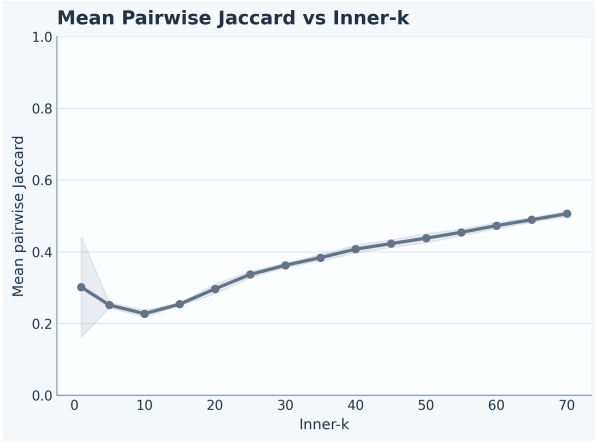


Figure 5: Mean pairwise Jaccard overlap by inner-k. Hard-set overlap increased with larger per-fold feature budgets.

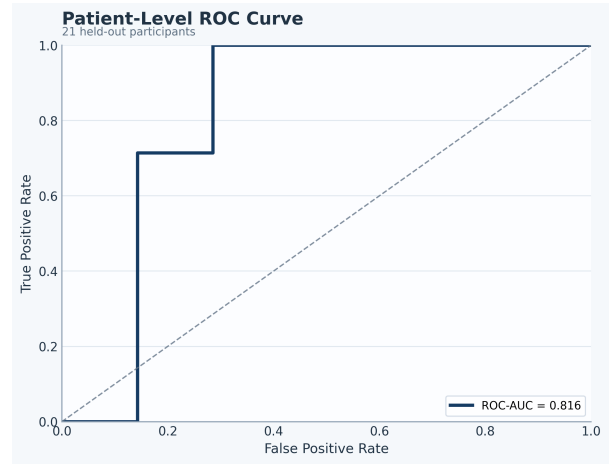


Figure 7: Patient-level ROC curve for the best hybrid run. The shaded region shows the 95% bootstrap confidence band.

sweep at inner-k=1. The linear SVM did not show the same pattern and reached its best ROC-AUC only at the broader inner-k=30 setting. Sparse per-fold selection therefore aligned with peak discrimination only for the more expressive model.

Figures 5 and 6 show a second related pattern. Increasing inner-k increased hard-set overlap and the total number of unique selected features. Larger feature budgets did not produce the strongest patient-level discrimination. The sweep concentrated its best hybrid performance at the sparsest end of the per-fold selection range. This pattern suggests that model capacity and per-fold sparsity mattered together rather than independently.

3.2. Best-run patient-level performance

The best hybrid configuration achieved patient-level ROC-AUC 0.816 [95% CI: 0.600–0.990], PR-AUC 0.579, balanced accuracy 0.786 [95% CI: 0.571–0.967], accuracy 0.810 [95% CI: 0.619–0.952], F1 0.714 [95% CI: 0.364–0.941], and MCC 0.571 over 21 held-out participants (Figure 7; Table 4). The permutation test

yielded a nominal raw $p = 0.008$ based on 500 iterations. Correcting for the structured sweep (2 model families) gives $p_{\text{corr}} = 0.016$, which survives at $\alpha = 0.05$; under the more conservative max-T correction across all 150 configurations, $p_{\text{corr}} = 0.302$ (Table 4 footnote). Bootstrap confidence intervals are wide, reflecting the small sample size of 21 participants. The PR-AUC of 0.579 indicates that the model retained useful precision-recall behavior in this cohort.

The corresponding participant-level confusion counts were 12 true negatives, 5 true positives, 2 false positives, and 2 false negatives. This error profile is consistent with the balanced-accuracy summary and shows that the strongest run retained useful sensitivity without sacrificing specificity. Figure 8 shows the same run in condensed metric form with 95% bootstrap confidence intervals.

The best SVM baseline reached balanced accuracy 0.643 and ROC-AUC 0.510 [95% CI: 0.119–0.881], with a non-significant permutation result (raw $p = 0.469$, uncorrected). This weaker performance came with a much larger per-fold feature budget (inner-k=30) and a far

Table 4

Best-performing configuration for each model family. Bootstrap confidence intervals are from 1,000 resamples at the 95% level.

Model	Window	Inner-k	ROC-AUC	PR-AUC	Bal. Acc.	F1	MCC	Feat./fold	Unique feat.	Mean Jaccard
Hybrid CNN-LSTM	6	1	0.816	0.579	0.786	0.714	0.571	1.0	7	0.190
Linear SVM	8	30	0.510	0.411	0.643	0.533	0.277	30.0	103	0.360

Bootstrap 95% CIs from 1,000 resamples. Hybrid permutation test: raw $p = 0.008$; primary correction (2 model families) $p_{\text{corr}} = 0.016$; robustness checks: max-T across 150 configurations $p_{\text{corr}} = 0.302$, full-sweep Bonferroni $p_{\text{corr}} = 1.00$.

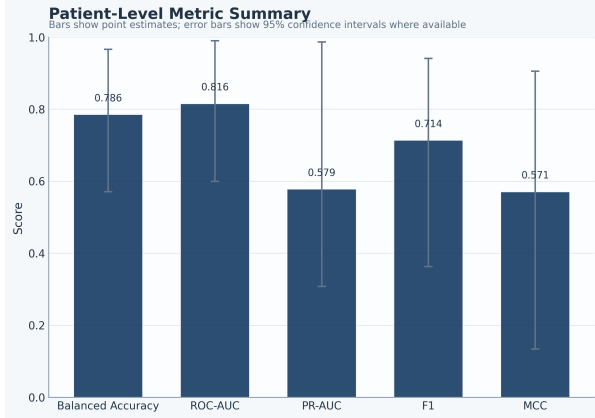


Figure 8: Patient-level metric summary for the best hybrid run. Error bars show 95% bootstrap confidence intervals.

larger overall candidate set. The contrast indicates that the sparse hybrid pipeline extracted stronger patient-level discrimination from a much smaller subset of features.

3.3. Intermediate baselines and ablation at the sparsest budget

To determine whether the hybrid architecture's advantage at $\text{inner-k}=1$ reflected generic benefits of non-linearity or specific architectural properties, we evaluated eight classical classifiers and two neural baselines under the identical configuration (6 s windows, $\text{inner-k}=1$, SMOTE, LOPO equalization, same dataset and folds). All ten baselines performed at or below chance (Table 5). The best classical model (Random Forest) reached ROC-AUC 0.296, compared with 0.816 for the hybrid. A 2-layer MLP (ROC-AUC 0.204) and a GAP variant of the hybrid architecture that removes self-attention and feature-pyramid fusion (ROC-AUC 0.235) both failed to exceed chance. No baseline achieved a significant permutation p -value (all $p \geq 0.9$).

An additional ablation removed SMOTE from the best hybrid configuration. Performance fell from ROC-AUC 0.816 to 0.582 (balanced accuracy 0.500, F1 0.308), indicating that SMOTE contributes substantially to the hybrid's performance in the 1-dimensional feature regime; this is consistent with the observation that synthetic oversampling provides the only mechanism for exposing class-boundary information when a single scalar feature is available.

A noise-injection ablation tested whether the hybrid architecture at $\text{inner-k}=1$ exploits the actual feature values

or relies on distributional artifacts accessible through batch statistics. After feature selection, all selected feature values were replaced with i.i.d. Gaussian noise ($\mathcal{N}(0, 1)$) and the model was retrained under the identical LOPO configuration. Performance fell to ROC-AUC 0.408 (balanced accuracy 0.393, F1 0.250), indistinguishable from chance. This confirms that the model relies on the actual feature values and is not driven by distributional cues.

These findings indicate that the hybrid CNN-LSTM's performance at extreme sparsity is not attributable to non-linearity alone, nor to SMOTE alone. The convolutional, recurrent, attention, and feature-pyramid stages, together with SMOTE-based training, appear jointly necessary to extract discriminatory signal from a single feature per fold.

The winning hybrid run selected an average of 1 feature per fold and yielded only 7 unique features that appeared in at least one fold across the entire participant-level validation process (Table 6). That degree of sparsity is one of the most distinctive findings in the sweep. Hard-set overlap was modest, with a mean pairwise Jaccard value of 0.190 and a mean Kuncheva index of 0.187. Figure 9 shows the small candidate pool, while the overlap statistics indicate that recurrence was concentrated in a few repeatedly reappearing features rather than in a broadly shared foldwise set.

A small set of features emerged repeatedly, though not in every fold, and the selected features were dominated by temporal and frontal-temporal descriptors. The most frequently selected feature was `tp10_spectral_entropy`, selected in 8 of 21 folds. The next most frequent were `left_frontal_temporal_diff_gamma`, `left_frontal_temporal_diff_beta`, and `af7_zero_crossings`. Three of the seven features listed in Table 6 appear in only 1 of 21 folds; the term "recurrent" should be interpreted with this sparsity in mind, and the sign-consistency values for these single-fold features carry limited interpretive weight.

3.4. Effect-direction consistency and class-conditional patterns

The recurrent hybrid biomarkers showed perfect mean sign consistency, with all recurrent features retaining the same effect direction across the folds in which they were selected. `tp10_spectral_entropy` and the frontal-temporal gamma and beta difference terms were repeatedly associated with the non-remission group, whereas `af7_zero_crossings` was more frequently selected in remission folds. The

Table 5

Intermediate baseline classifiers and ablations evaluated at the best hybrid configuration (6 s, inner-k=1). All models used identical data, feature selection, and validation splits. Neural baselines and ablation trained on NVIDIA RTX 3090.

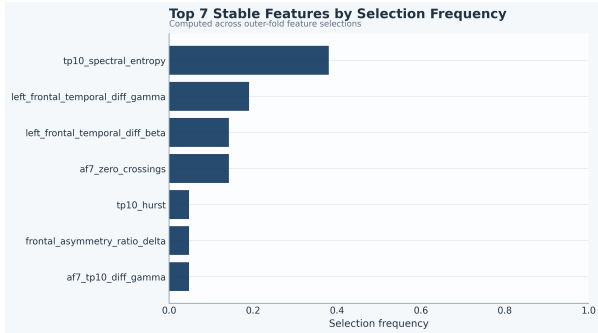
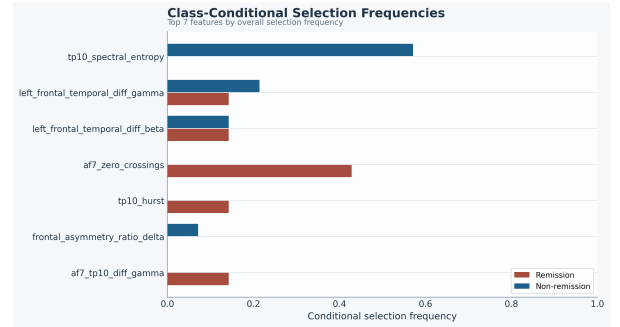
Model	ROC-AUC	Bal. Acc.	MCC	F1	p (perm.)	Notes
Hybrid CNN-LSTM (best)	0.816	0.786	0.571	0.714	0.008	Full architecture, SMOTE
<i>Classical models (inner-k=1, SMOTE)</i>						
Random Forest	0.296	0.321	-0.337	0.222	0.930	
Extra Trees	0.286	0.500	0.000	0.375	0.934	
AdaBoost	0.286	0.393	-0.204	0.250	0.942	
Gradient Boosting	0.276	0.393	-0.204	0.250	0.936	
Decision Tree	0.276	0.500	0.000	0.308	0.946	
KNN	0.255	0.464	-0.067	0.353	0.974	
SVM-RBF	0.204	0.429	-0.135	0.333	0.978	
SVM-Linear	0.204	0.357	-0.277	0.300	0.988	
<i>Neural baselines (inner-k=1, SMOTE)</i>						
MLP (2-layer)	0.204	0.429	-0.135	0.333	>0.95	
GAP variant (no attn.)	0.235	0.464	-0.067	0.353	>0.95	
<i>Ablation (hybrid architecture)</i>						
Hybrid, no SMOTE	0.582	0.500	0.000	0.308	>0.05	SMOTE essential at $k=1$
<i>Sanity check</i>						
Hybrid, noise-injected features	0.408	0.393	-0.207	0.250	>0.50	Selected features replaced with $\mathcal{N}(0, 1)$

All models evaluated at window size 6 s, inner-k=1, SelectKBest (F -classif), LOPO group equalization enabled, patient-level mean aggregation. The hybrid result is from the original 150-run sweep (reference). Classical models run on CPU; neural baselines and ablation run on NVIDIA RTX 3090. GAP = Global Average Pooling variant removing self-attention and feature-pyramid fusion.

Table 6

Features from the best hybrid run that appeared in at least one fold. Δ selection is remission minus non-remission selection share. Sign consistency is 1.000 for features appearing in multiple folds; single-fold entries should be interpreted with caution.

Feature	Folds	Share	Δ sel.	Sign cons.	Mean d	Effect direction
tp10_spectral_entropy	8	0.381	-0.571	1.000	-0.935	Non-remission > remission
left_frontal_temporal_diff_gamma	4	0.190	-0.071	1.000	-1.059	Non-remission > remission
left_frontal_temporal_diff_beta	3	0.143	0.000	1.000	-0.915	Non-remission > remission
af7_zero_crossings	3	0.143	0.429	1.000	0.814	Remission > non-remission
tp10_hurst	1	0.048	0.143	1.000	-1.010	Non-remission > remission
af7_tp10_diff_gamma	1	0.048	0.143	1.000	-1.203	Non-remission > remission
frontal_asymmetry_ratio_delta	1	0.048	-0.071	1.000	0.794	Remission > non-remission

**Figure 9:** Selection-frequency ranking for the features identified from the best hybrid run.**Figure 10:** Class-conditional selection frequencies for the recurrent biomarkers from the best hybrid run. Remission is shown in red and non-remission in blue.

class-conditional selection frequencies make that separation visible at the level of foldwise recurrence (Figure 10).

The effect-size summaries followed the same broad pattern, with af7_zero_crossings retaining a positive mean effect size and the main TP10 and frontal-temporal

features retaining negative mean effects (Figure 11). This distinction matters clinically. The selection-frequency deltas do not justify definitive mechanistic claims. They provide a concrete interpretation target for follow-up

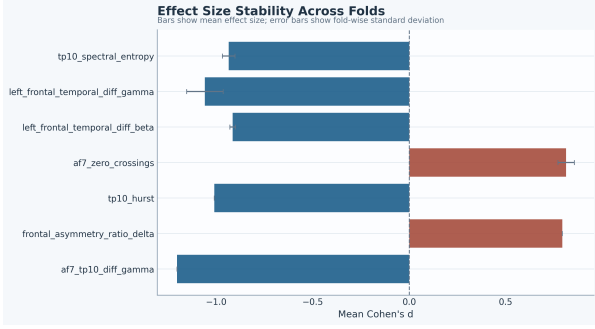


Figure 11: Effect-size stability for the recurrent biomarkers from the best hybrid run. Positive values indicate remission-skewed effects and negative values indicate non-remission-skewed effects.

validation. Taken together, these results indicate that useful patient-level discrimination can be achieved with an extremely sparse feature budget, while the recurrent biomarkers retain interpretable directionality even when fold-to-fold set overlap remains modest.

3.5. Aggregation robustness

Patient-level metrics in the main analysis were derived by averaging window probabilities within each participant. To assess sensitivity to outlier windows, we recomputed the best hybrid run's patient-level metrics using median aggregation and 25% trimmed-mean aggregation. Median aggregation yielded ROC-AUC 0.857, PR-AUC 0.620, and MCC 0.571; trimmed-mean aggregation yielded ROC-AUC 0.837 and MCC 0.571. Both alternative aggregation methods matched or exceeded the original mean-aggregated performance (ROC-AUC 0.816), confirming that the main results are not driven by outlier windows and that the reported mean-based metrics are conservative rather than inflated.

4. Discussion

The strongest patient-level discrimination in this sweep emerged from the sparsest tested configuration. The best hybrid run used $\text{inner-k}=1$, selected only 7 unique features appearing in at least one fold, and exceeded the best linear baseline by a large margin on ROC-AUC and MCC. Figure 4 also shows that this was not a generic effect of smaller feature budgets across both models. In the low-budget region, the hybrid model retained stronger ROC-AUC than the linear SVM and reached its peak performance at the sparsest tested setting, whereas the linear model reached its best ROC-AUC only after a broader per-fold budget. In this cohort, the value of the more expressive model and the value of strong sparsity appear to have operated together.

Statistical inference was assessed with permutation tests and three tiers of multiple-comparisons correction to account for the structured sweep design. The raw permutation p-value of $p = 0.008$ survives the primary structured correction (2 model families, $p_{\text{corr}} = 0.016$) at $\alpha = 0.05$. As robustness checks, we computed a max-T

permutation correction across all 150 configurations ($p_{\text{corr}} = 0.302$) and a full-sweep Bonferroni ($p_{\text{corr}} = 1.00$), the latter being mechanically uninformative given the 150-test multiplier. We adopt the 2-family correction as the primary inference because the inner-k and window-size factors represent exploratory levels of a single hypothesis rather than independent comparisons, and report all tiers for complete transparency. Several additional caveats qualify the results. The study is limited to $n = 21$ participants with a 7:14 class imbalance, producing wide bootstrap confidence intervals (e.g., ROC-AUC 95% CI: 0.600–0.990) that span both chance-level and near-perfect performance. The hybrid model's hyperparameters were inherited from a standard configuration rather than tuned specifically on this cohort; while this reduces researcher degrees of freedom, the configuration may not be optimal for the present data.

One plausible explanation for the performance gap between models is that aggressive per-fold pruning exposed a compact signal set whose interactions were more useful to the hybrid architecture than to a linear decision boundary. Supporting this interpretation, eight classical classifiers and two neural baselines (2-layer MLP, GAP architecture variant) were evaluated at the identical $\text{inner-k}=1$ configuration; all performed at or below chance (ROC-AUC 0.204–0.296, Table 5). Removing SMOTE from the hybrid reduced ROC-AUC from 0.816 to 0.582, and replacing all selected features with $\mathcal{N}(0, 1)$ noise reduced it further to 0.408. Together these results show that the full CNN-BiLSTM-attention pipeline with SMOTE-based training is jointly required: neither non-linearity alone, nor SMOTE alone, nor a simplified deep architecture suffices, and the model relies on the actual feature values rather than distributional cues.

We did not compute formal paired statistical tests (e.g., DeLong or bootstrapped AUC comparison) between the two model families because the best configurations operate at different inner-k values (1 vs. 30). At matched inner-k values near the hybrid's optimum, the SVM operates near chance (ROC-AUC ≈ 0.4 –0.5; Figure 4). The intermediate-baseline experiment (Table 5) provides complementary evidence: ten alternative models all fail at $\text{inner-k}=1$, supporting the conclusion that the hybrid's advantage is not a statistical artifact of the sweep. This interpretation remains provisional but is consistent with the sweep pattern and the marked separation between the best sparse hybrid run and the broader-budget linear baseline.

This result fits part of the EEG depression literature and supports a measured interpretation. Prior work has argued that EEG can provide clinically useful baseline markers in major depressive disorder and treatment-response settings (Leuchter et al., 2010; Olbrich and Arns, 2013; Olbrich et al., 2015). Related discussions have also emphasized tighter biomarker discipline, better standardization, and more realistic expectations about generalization (Fu et al., 2019; Moncy et al., 2025; Simmatis et al., 2023). Machine learning studies have likewise reported encouraging treatment-response signals, including combinations of

Table 7

Comparison with selected EEG-based treatment-response prediction studies in depression.

Study	Method	EEG channels	<i>n</i>	AUC	Ext. val.	Task
Moncy et al. (2024)	CNN, PSD features	4*	21	0.76 [†]	No	tDCS remission
Xiao et al. (2025)	PLV + ML	4*	21	0.72 [†]	No	tDCS remission
Al-Kaysi et al. (2017)	SVM, RF, LR	19	80	0.82	No	Antidepressant
Jaworska et al. (2019)	SVM, LR	32	73	0.83	No	Antidepressant
Oakley et al. (2022)	LORETA + SVM	72	65	0.75	No	Sertraline/placebo
Ebrahimzadeh et al. (2021)	Nonlinear EEG	19	30	0.90	No	Antidepressant
Shalbaf et al. (2018)	Nonlinear + SVM	19	45	0.88	No	rTMS
Zhdanov et al. (2020)	SVM, spectral	19	44	0.76	No	Escitalopram
This study	CNN-LSTM, sparse	4*	21	0.82	No	tDCS remission

*Low-montage 4-channel headset (AF7, AF8, TP9, TP10). [†]Estimated from reported metrics. AUC = area under the ROC curve (patient-level where available). Ext. val. = external validation cohort. Studies selected for relevance to EEG-based treatment response prediction in depression; the list is illustrative rather than exhaustive.

EEG and clinical covariates (Al-Kaysi et al., 2017; Jaworska et al., 2019; Oakley et al., 2022). Other modelling studies reach similar conclusions while still exposing limits from data heterogeneity, cohort size, or insufficient interpretability (Ebrahimzadeh et al., 2021; Shalbaf et al., 2018; Zhdanov et al., 2020). Recent predictive overviews and reviews continue to highlight the same translational tension (Cohen et al., 2023; Watts et al., 2022). Table 7 places the present results in the context of comparable EEG-based treatment-response prediction studies. The achieved ROC-AUC of 0.816 is competitive with prior work using larger montages and cohorts, despite the present study's substantially reduced channel count and sparse feature budget. However, direct numerical comparison is complicated by differences in clinical populations, outcome definitions, and validation methodology; the table is provided for contextual orientation rather than formal benchmarking.

The present sweep adds a feature-selection view in which larger feature budgets yield more overlap and many more unique selected features. Those larger budgets do not improve discrimination. The panels support a small set of frequently selected features with consistent direction of class separation. That reading keeps the modest Jaccard and Kuncheva scores in view and gives appropriate weight to the stronger effect-direction signal. However, the low Jaccard (0.190) and Kuncheva (0.187) scores indicate that the selected feature changes in most folds, which is a limitation for biomarker reproducibility claims. Three of the seven features appear in only a single fold, and the “perfect” sign consistency values for those entries should be interpreted cautiously. The Jaccard increase with larger inner-*k* is partly mechanical (selecting *k* features from a fixed pool of 247 will produce more overlap by construction as *k* increases), so this trend should not be over-interpreted as evidence of underlying signal stability. Methodologically, this framing matters because the depression EEG biomarker literature still contains substantial publication bias, limited out-of-sample validation, and known vulnerability to leakage from pre-cross-validation feature processing (Shim et al., 2021;

Widge et al., 2019). The present pipeline addresses that problem by embedding selection and rebalancing inside participant-level leave-one-participant-out folds and then summarizing recurrence and directionality instead of reporting only a single global feature list (Shen et al., 2025).

The same clinical program has also supported other signal representations in recent PSD-based and PLV-based studies, which located informative response-related signal in low-density temporal and frontal-temporal patterns (Moncy et al., 2024; Xiao et al., 2025b). In the present sweep, the most frequently selected features again concentrated in TP10-derived and frontal-temporal descriptors. That regional convergence supports the view that the low-montage recordings may contain response-relevant information across multiple feature families. However, the current study provides only within-cohort evidence; the generalizability of these features under different montage configurations, clinical populations, and treatment protocols remains untested.

The specific features selected in the present sweep are neurophysiologically plausible. Elevated temporal spectral entropy in non-remission participants suggests greater irregularity or complexity in the resting-state signal at the TP10 site, which overlies the temporoparietal junction, a region implicated in the default mode network and self-referential processing, both of which are dysregulated in depression. Higher entropy may reflect less organized or more variable neural dynamics that are less amenable to tDCS-induced modulation. The frontal-temporal beta and gamma difference features point toward inter-regional communication in higher frequency bands, which have been linked to top-down cognitive control and attentional processes mediated by fronto-temporal networks. tDCS applied over the dorsolateral prefrontal cortex (approximated by AF7/AF8) is thought to modulate excitability in these circuits, and baseline differences in frontal-temporal coupling may index a network's capacity to respond to such modulation. The AF7 zero-crossing count, a time-domain measure of signal volatility, showed the clearest remission-associated pattern and may capture a more dynamic or responsive frontal signal state that

is conducive to stimulation-induced plasticity. These interpretations are necessarily speculative given the study's sample size and lack of independent replication, but they provide concrete, testable hypotheses for future mechanistic work.

Related tDCS studies in bipolar depression from the same group point in a similar low-montage direction. PSD-based modeling again highlighted AF7 and TP10 signal content, and PLV analyses emphasized frontal-temporal and temporoparietal synchronization patterns (Moncy et al., 2025; Xiao et al., 2025a). These bipolar studies address a different diagnosis and treatment protocol. They support reduced-montage feasibility and signal plausibility in adjacent affective cohorts.

This study has several limitations. First, the relatively small participant size ($n = 21$) with a 7:14 class imbalance limits statistical power and generalizability; the wide bootstrap confidence intervals (e.g., ROC-AUC 95% CI: 0.600–0.990) reflect this uncertainty. Although SMOTE and group equalization were applied within training folds, the fundamental class imbalance remains a constraint. Second, the primary corrected p -value ($p_{\text{corr}} = 0.016$) survives the structured 2-family Bonferroni correction but does not survive the more conservative max-T correction across all 150 configurations ($p_{\text{corr}} = 0.302$); replication in an independent cohort is needed. Third, there is no external validation cohort, so the reported performance and identified features cannot yet be assessed for generalizability to independent populations, alternative low-density montages, or different clinical settings. Fourth, important clinical variables such as medication status, illness duration, episode count, and psychiatric comorbidities were not available; these may influence both EEG features and treatment outcomes. Fifth, the 4-channel low-density montage trades spatial coverage for practical deployability. Recording only AF7, AF8, TP9, and TP10 captures frontal and temporal regions while omitting central, parietal, and occipital sites. This limits source-localization precision and precludes analysis of well-studied posterior alpha asymmetry markers. However, the reduced setup also enables a fully home-based, self-administered protocol that would be impractical with a 19- or 32-channel research montage, and the observed signal in these four channels supports the feasibility of this trade-off for remote tDCS monitoring. Whether a richer montage would yield higher predictive performance remains an open question for future head-to-head studies. Sixth, the feature stability metrics (mean Jaccard 0.190, Kuncheva 0.187) indicate limited cross-fold consistency; however, median-based patient aggregation improved ROC-AUC to 0.857, suggesting that the feature signal is stable even when fold-to-fold feature identity varies. Seventh, when inner- $k=1$, SMOTE operates in a 1-dimensional feature space where the standard rationale for synthetic minority oversampling does not fully apply. The no-SMOTE ablation confirmed this concern: removing SMOTE reduced the hybrid's ROC-AUC from 0.816 to 0.582, so SMOTE is empirically essential at this

setting despite the theoretical caveat. Eighth, the study does not benchmark against recent transformer-based or graph-neural-network EEG classifiers, which represent an active area of development. Ninth, the hybrid architecture contains approximately 1.2 million parameters and requires GPU acceleration for training; inference time and memory footprint, while modest for a research server, may constrain deployment on edge devices. Tenth, the SOTA comparison (Table 7) is contextual rather than a formal benchmark due to differences in clinical populations, outcome definitions, and validation methodology between studies; direct method-level comparisons would require harmonized datasets and protocols.

The recurring features can be treated as plausible candidate features from this cohort that now warrant follow-up validation rather than as validated clinical biomarkers. Future work should test whether the same sparse feature set recurs under independent cohorts, alternative low-density montages, and explicit external validation (Dagnino et al., 2023). Adjacent response-prediction work in depression and neuromodulation provides useful targets for that next stage (Bailey et al., 2023; Funk and George, 2008; Guo et al., 2024). Home-based and transport-oriented studies offer one clear direction for follow-up (Kratter et al., 2026; Romero Marin et al., 2026; Waller et al., 2026). More recent response-signature studies also provide relevant benchmarks for that next phase (Reissmann et al., 2026; Sheen et al., 2024; Zeidabadi and Rashidi, 2025). Related protocol and review work in treatment-resistant depression can help frame that validation effort (Pettorruso et al., 2024; Ulrich et al., 2025). These extensions should keep stringent patient-level evaluation and sparse interpretable feature budgets in the core study design, ideally with pre-registered analysis plans to limit researcher degrees of freedom.

5. Conclusion

This study demonstrates that low-montage resting-state EEG can support patient-level prediction of remission outcomes in a home-based tDCS trial using an extremely sparse feature-selection framework. The strongest model achieved promising discrimination (ROC-AUC 0.816, MCC 0.571) while selecting only a small set of features, though the result does not survive correction for the full sweep of 150 configurations and should be considered exploratory.

Rather than relying on high-dimensional feature spaces, the results suggest that compact feature candidates centered on temporal entropy and frontal-temporal interactions may capture clinically relevant signal. These findings extend prior PSD-based analyses in the same cohort and support a convergent regional signal interpretation. However, the limited sample size ($n = 21$), absence of an external validation cohort, and modest feature stability (Jaccard 0.190) indicate that these features should be treated as hypothesis-generating rather than as validated clinical biomarkers. Prospective validation in independent cohorts

with pre-registered analysis plans is necessary before any clinical application can be considered.

CRediT authorship contribution statement

Abin Daniel Zorto: Writing - review & editing, writing - original draft, visualization, validation, software, methodology, formal analysis, data curation, conceptualization.. **Jijomon C. Moncy:** Writing - review & editing, writing - original draft, visualization, validation, software, resources, methodology, formal analysis, data curation, conceptualization.. **Wenyi Xiao:** Writing - review & editing, writing - original draft, visualization, validation, software, resources, methodology, formal analysis, data curation, conceptualization.. **Mhd Saeed Sharif:** Writing - review & editing, writing - original draft, validation, supervision, resources, project administration, methodology, investigation, funding acquisition, formal analysis, conceptualization.. **Cynthia H.Y. Fu:** Writing - review & editing, writing - original draft, validation, supervision, resources, project administration, methodology, investigation, funding acquisition, formal analysis, conceptualization..

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